Sahaj Bhadja

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Summary

Full-Stack Software Engineer with 2+ years of experience building **Python backends**, **TypeScript/React interfaces**, high-volume databases, and **automated verification pipelines** for reliable production tools.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Object-Oriented Programming

Experience

USC Dornsife | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Achieved 99%+ reliability across 5K+ monthly jobs through **data validation and reconciliation**, error handling, cloud storage, CI/CD releases, and stable production monitoring for dependable internal workflows reliably.
- Reduced dashboard latency from 1.2s to 350ms through **SQL indexing and query optimization**, designing normalized schemas and row-level isolation for a multi-tenant analytics platform supporting 500+ users.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.
- Processed 1K+ files per batch using **Python multithreading**, validating heterogeneous inputs while isolating failures and preserving responsive user workflows during concurrent processing and sustained production traffic reliably.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) | *India*
- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while debugging integrations across Windows, Linux, and macOS production environments reliably at scale.
- Automated builds and 20+ unit/integration suites through **Jenkins, GitHub, PyTest, and Mockito**, generating failure reports and coverage summaries across Linux, Windows, and macOS with repeatable nightly workflows.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) | *India*
- Synchronized 2M+ records through an eight-service **React and Node.js platform** covering authentication, resources, customers, billing, leads, permissions, and notifications with WebSockets and Azure SQL across locations and workflows.
- Reduced resource wastage by 42% through **automated testing and deployment**, implementing JWT access controls, WebSocket updates, Azure monitoring, Docker/Jenkins pipelines, and ARM templates across eight modular service domains.

LST Capitals | *Web Developer Intern* | *India* **November 2022 – January 2023**
- Delivered **sub-200ms Flask APIs** across 50K+ booking records, implementing pricing, validation, payment webhooks, Firebase synchronization, and database transactions while preventing conflicting reservations across concurrent customer workflows reliably.
- Improved cross-platform compatibility by 15% through **React Native and TypeScript interfaces**, integrating Android, iOS, and web booking workflows while reducing design errors by 25% through reusable components.

Skills

Languages: Python, TypeScript, JavaScript, Java, C++, SQL, Bash
Frontend & Backend: React, React Native, Node.js, FastAPI, Flask, Express.js, REST APIs, WebSockets, Microservices
Data & Platforms: PostgreSQL, MySQL, MongoDB, Azure SQL, AWS RDS, Firebase, Redis, Data Modeling, Query Optimization
Cloud & Delivery: AWS, Azure, Docker, Kubernetes, Jenkins, Git, GitHub, CI/CD, ARM Templates, Infrastructure as Code
Quality & Reliability: Unit Testing, Integration Testing, Automated Testing, Regression Testing, PyTest, Mockito, Jest, Selenium, Debugging, Monitoring, Error Handling, Data Validation

Projects

Mission Operations Console - Design Concept
- Designed a **mission-operations console** concept with TypeScript dashboards, Python services, command validation, role-based access, telemetry history, health monitoring, audit logs, and operator-confirmed workflows for dependable rover support.

Hardware Verification Data Platform - Design Concept
- Designed a **hardware-verification platform** concept ingesting test runs, measurements, artifacts, and failures into searchable relational records with automated regression checks, traceable approvals, deployment gates, and technician-facing dashboards.

CPU-Scheduling Simulator | Live
- Built a **CPU-scheduling simulator** implementing FCFS, SJF, SRTF, Priority, and Round Robin algorithms with configurable workloads, timeline visualization, comparative metrics, and deterministic validation across scheduling scenarios reliably.